QuecPython PH-7(C4-P02)

Core board Specification and User Guide

LPWA&LTE Standard Series

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Quectel Wireless Solutions Co., Ltd.
Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China
Tel: +86 21 5108 6236
Email: info@quectel.com

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About the Document

Revision History

Version	Date	Author	Description
1.0.0	2024-11-14	Chavis CHEN	Creation of the document

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1 Introduction

The PH-7 core board adopts a 2.54mm pin design, providing 38 pin pins, including functional pins and enable control pins. Mainly aimed at providing support and convenience for customer development, debugging, and mass production equipment.

Design concept: PH-7 is neutral, water is the most common neutral substance, widely, closely related to daily life, and indispensable; Secondly, it also means that users should be cautious and have a soft mindset like water. Be water , My friend.

The core board supports Python and Open secondary development, and this manual mainly introduces the Python version.
Python documents: https://python.quectel.com/download
Python development API reference：https://python.quectel.com/doc/API_reference/zh/index.html

1.1. Applicable Core Board and Model

The development boards and updated module models applicable to this manual are as follows:
Table 1: Applicable Core Board

Development board model	Description
QuecPython BG&EG Core Board	Pin type core board, leading out commonly used and critical pins

Table 2: Applicable Model

Development board model	Description
QuecPython BG&EG Core Board	Applicable to the entire BG95, BG96 series, EG915UEUAB, EG915ULAAB, EG912UGLAA and other EG series, including but not limited to the above module models. There may be some shortcomings in the adaptation of certain functional applications. For details, please refer to the Python development API documentation on the QuecPython official website

1.2. Features

- Complete functional coverage
- Compatible with various module models
- Compatible with different voltage levels

- Pin method interface, convenient for external debugging
- support GPIO/UART/IIC/SPI/USB
- Support lithium battery charging function
- EG supports analog audio input and output
- support Python development or Open C development

2 Top and Bottom Views

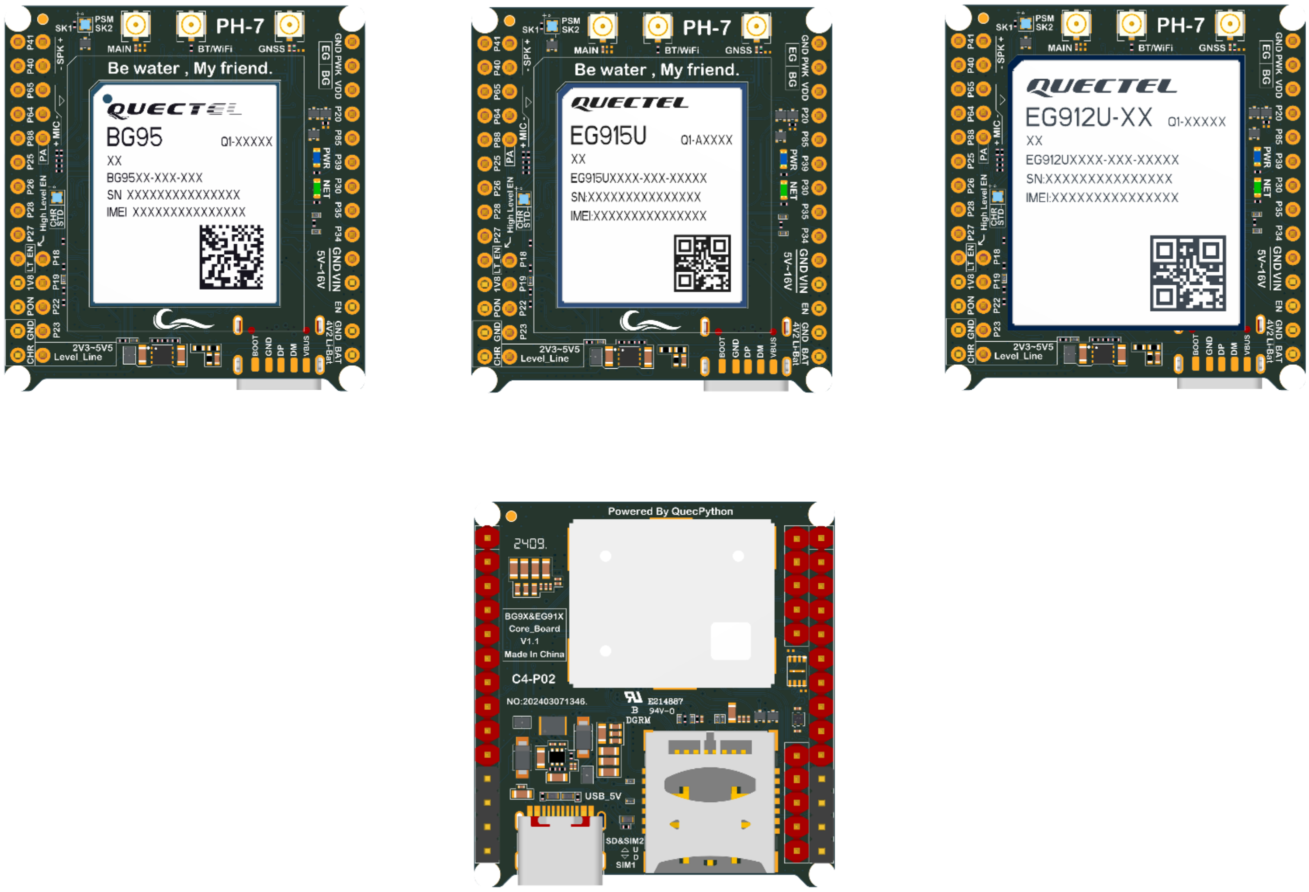


Figure 1: PH-7(C4-P02) Core Board View

3 Specifications

This chapter mainly outlines the specific specifications and parameters of the PH-7 core board, as well as some device precautions.

Table 3: Specifications

	条目	描述
Electrical parameters	temperature	operation temperature: -35 ~ 75℃ (See Note 1 below) Extended temperature: -40 ~ 85℃ (See Note 1 below)
	operation temperature	Unknown, according to the actual situation.
	Power supply interface	2.54mmcontact pin、Type-C (USB2.0/USB3.0)
	Supply Voltage	Normal operating voltage range for USB power supply: ≤5.1V:3A; ≥5V:2A External pin power supply range:≤16V:3A； ≥5V:2A BAT battery power supply: 3.7V rechargeable lithium battery
RF Performance	Band	Please refer to the module specification sheet for details
		Please refer to the module specification sheet for details
	Transmission power	Please refer to the module specification sheet for details
		Please refer to the module specification sheet for details
UART	Interface type	Including functional reuse, please refer to the reuse table for details. Connection method: Pin (2.54mm)
SPI	Interface type	Including functional reuse, connection method: pin (2.54mm)
IIC	Interface type	Including functional reuse, connection method: pin (2.54mm)
USB*1	USB Interface	Type-C female head, supporting power supply, software debugging, firmware burning, and updating Python scripts
LED*7	PWR	Power indicator, ice blue, always on after power on
	NET	The default is the module network status indicator light. Connect module P21 pin
	SK1	The onboard SIM1 card slot indicator is on when SIM card 1 is inserted and recognized normally, and off when SIM card 1 is exited.

	SK2	The onboard SIM2 card slot indicator lights up when SIM card 2 is inserted and recognized normally, and turns off when SIM card 2 is exited. (Only supports EG dual card series, firmware support required.)
	PSM	When the module turns on PSM or enters PSM state, it stays on for a long time.
	CHR	Blue, charging duration on
	STD	Emerald green, full of long-lasting brightness
ADC	Interface type	ADC0 connects to VBAT by default, meeting VBAT: ADC0=4:1
GPIO	Interface type	Including multiplexed GPIO, please refer to the functional reuse table for details, 2.54mm pin type external connection
Level_Line	Level_Line	2.3V-5.5V, level adaptive, used for level conversion IC host side level. If the default level is 3.3V, it can be disconnected
PWM	Interface type	Including functional reuse, 2.54mm pin type external connection
LT_EN	Level Enable	active high
PON	PSM Wakeup	PSM-EINT pin, can wake up the module from PSM
MIC&SPK	Analog audio	Analog audio interface
PA	PA Enable	Pull up to enable SPK PA output

Table 4: Turn-on and BOOT

Auto Turn-on	EG	EG series short circuiting pins automatically power on
	BG	BG series short circuiting pins automatically power on
BOOT	USB_BOOT	Introduce in the form of test points

Note

1. The scope of work and equipment performance meet the requirements of 3GPP standards.
2. In this scope of work, RF and network are basically not affected, with only a few indicators exceeding 3GPP standards. When the working temperature is restored, all indicators still comply with 3GPP standards.

4 Pin Descriptions

This section introduces the pins and sockets of the PH-7 core board, and the functional description is as follows:

4.1. Pin Views

Taking the EG915U core board as an example:

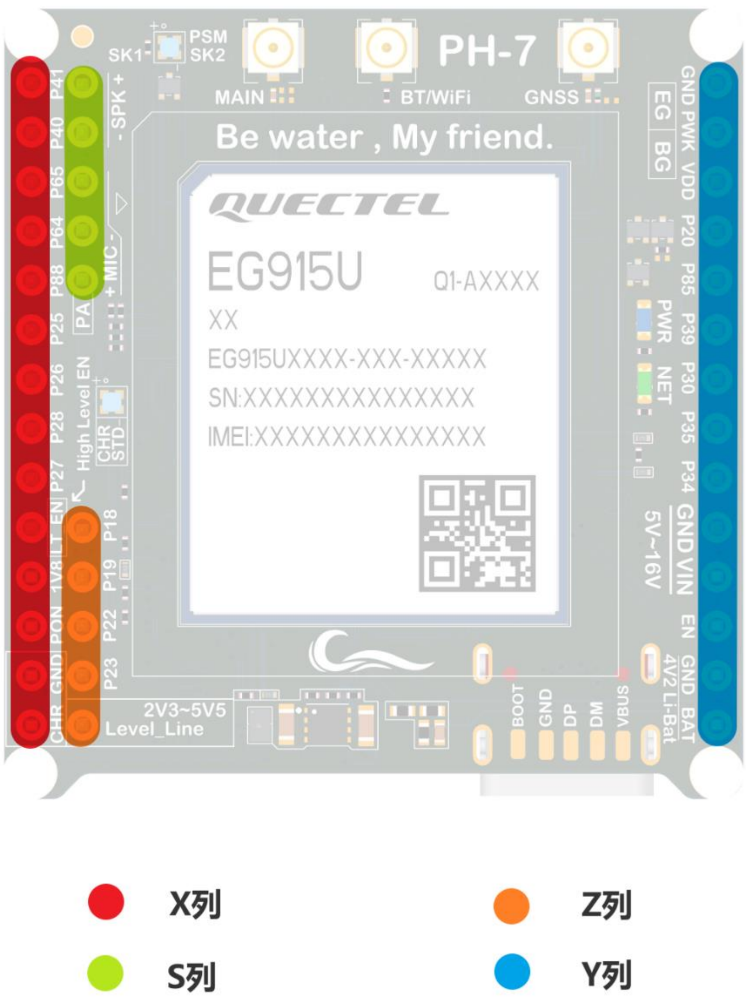


Figure 2: Core Board Pin View

Table 5: Core board pin function reuse table

GPIO14		BG_CS1		BG&EG_SDA1	**P41**	X1	S1	**SPK_N**	Y1	**GND**					
GPIO13		BG_CLK1		BG&EG_SCL1	**P40**	X2	S2	**SPK_P**	Y2	**POWERKEY**					
BG_GPIO16	BG_RX2	BG_MISO1			**P65**	X3	S3	**MIC_N**	Y3	**VDD_EXT**					
GPIO15	BG_TX2	BG_MOSI1		EG_MOSI0	**P64**	X4	S4	**MIC_P**	Y4	**P20**			EG_PWM0	BG_GPIO22	EG_GPIO16
GPIO21				EG_MISO0	**P88**	X5	S5	**PA**	Y5	**P85**		BG_PWM1		BG_GPIO18	EG_GPIO18
GPIO9		BG_SDA2	BG_CS0	EG_CS0	**P25**	X6			Y6	**P39**	RI			BG_GPIO30	EG_GPIO25
GPIO10		BG_SCL2	BG_CLK0	EG_CLK0	**P26**	X7			Y7	**P30**	DTR			BG_GPIO24	EG_GPIO20
GPIO12	BG&EG_RX1		BG_MISO0		**P28**	X8			Y8	**P35**	TX	BG_UART4	EG_UART2	BG_GPIO26	
GPIO11	BG&EG_TX1		BG_MOSI0		**P27**	X9			Y9	**P34**	RX			BG_GPIO25	
					LT_EN	X10	Z1	**P18**	Y10	**GND**					
					1V8	X11	Z2	**P19**	Y11	**VIN**					
					PON	X12	Z3	**P22**	Y12	**EN**					
					GND	X13	Z4	**P23**	Y13	**GND**					
					CHRG	X14	Z5	**Level_Line**	Y14	**BAT**					
							Z1			BG_SCL0		BG_CLK2	EG_RX4	BG_GPIO5	EG_GPIO5
							Z2			BG_SDA0		BG_CS2	EG_TX4	BG_GPIO6	EG_GPIO6
							Z3	DBG_RX		BG_RX0		BG_MISO2		BG_GPIO7	
							Z4	DBG_TX		BG_TX0		BG_MOSI2		BG_GPIO8	

Note

BG: Suitable for BG9X model module

EG: Suitable for EG91X model module

Basic serial port function pins, compatible with AT standard firmware

*The above table does not represent the full range of pin functions. Please compare according to the specific model used

4.2. Function Description

For the convenience of users to quickly get started operating the core board, this section briefly introduces the usage methods and precautions of the core board functions.

4.2.1. Charing and USB

When inserting the Type-C interface to connect to USB, the charging function is enabled by default and cannot be turned off. If the charging function is not used, there is no need to pay attention.
If using an external power charging method, please connect the power ground to the positive pole of the 5V-16V power supply to the X13 X14 pin to charge the battery mounted on the Y13 Y14 pin. (Please make sure to mount a 4.2V rechargeable battery)
It is not recommended to use USB while connecting CHRG (X14 pin) for charging.

4.2.2. Power Enable

The core board is enabled by default when powered on. If the core board needs to be powered off during the use of the entire device.
1. External direct disconnection of VIN power supply
2. Pull down the EN (Y12) pin to turn off the DCDC output. (See Note 1 below)

4.2.3. Level Translation

The core board is equipped with a 3.3V level conversion IC. To reduce power consumption, the level conversion is set to an unactivated state by default. During use, please externally pull up the LT-IN (Y10) pin, or short-circuit the adjacent two pins LT-IN (Y10) and 1V8 (Y11) to enable level conversion.

4.2.4. GNSS

The GNSS function depends on the module model. Taking BG95 as an example, the core board defaults to using active antennas and has already enabled power supply to the active antennas by default.
1. AT standard firmware, use AT+QPGS=1 to enable GNSS function. No need to control active power supply.
2. Secondary firmware development, due to the unstable initial voltage level of the pins, if GNSS active power supply needs to be turned on, please lower the P37 pin of the module. Pull up pin P37 to turn off GNSS active power supply, SD and SIM
Taking EG912UGLAA as an example, if SD storage function is required, after inserting the SD card, please first raise the P36 pin to enable power supply to the SD, and then use the SD function. Due to the overlapping position of SIM2 and SD card, if the firmware supports SIM2, SIM2 and SD cannot be used simultaneously.

4.2.5. Flash (optional)

On board Flash is not pasted by default. If you need to use Flash, please contact us separately.

4.2.6. Analogue Audio

The core board carries MIC circuit and SPK&PA amplifier circuit, taking EG912UGLAA as an example.
1. When using AT standard firmware, if external audio playback is required, please raise the PA (S5) pin to enable PA
2. When using Python firmware, simply pull up the PA (S5) externally or pull up the P114 pin in the module program (see Note 2 below)

4.2.7. ADC

The core board is equipped with an ADC for developers to use. ADC0 is connected to VBAT by default, meeting the VBAT: ADC0=4:1 requirement.

4.2.8. Low Power

For module models that support PSM, the power consumption of the core board can be reduced to an extremely low level. For specific implementation, please refer to the module specification book or QuecPython official website.

4.2.9. PSM and Wakeup

For module models that support PSM, the core board can externally wake up modules that enter low-power PSM.
1. Lower the POWERKEY.
2. Pull up the PON (X12) pin, or short-circuit adjacent pins X12 X11. (See Note 3 below)备注

1. The high level of EN depends on the VIN voltage. For example, if VIN is 12V, then the EN level is 12V. If external EN is pulled down, it is necessary to use MOS transistor control and not directly connect IO of different levels for control.
2. As the PA pin is directly connected to the module pin with a voltage level of 1.8V, when the external pull is high, please connect the Y11 or X3 pin of the core board.
3. As PON is a direct connection module pin with a voltage level of 1.8V, when the external pull is high, the connection is directly short circuited to the X12 X11 pins of the core board

5 Antenna Interface

The core board is equipped with RF connectors (sockets) for easy antenna connection. The size of the antenna connector is shown in the following figure.

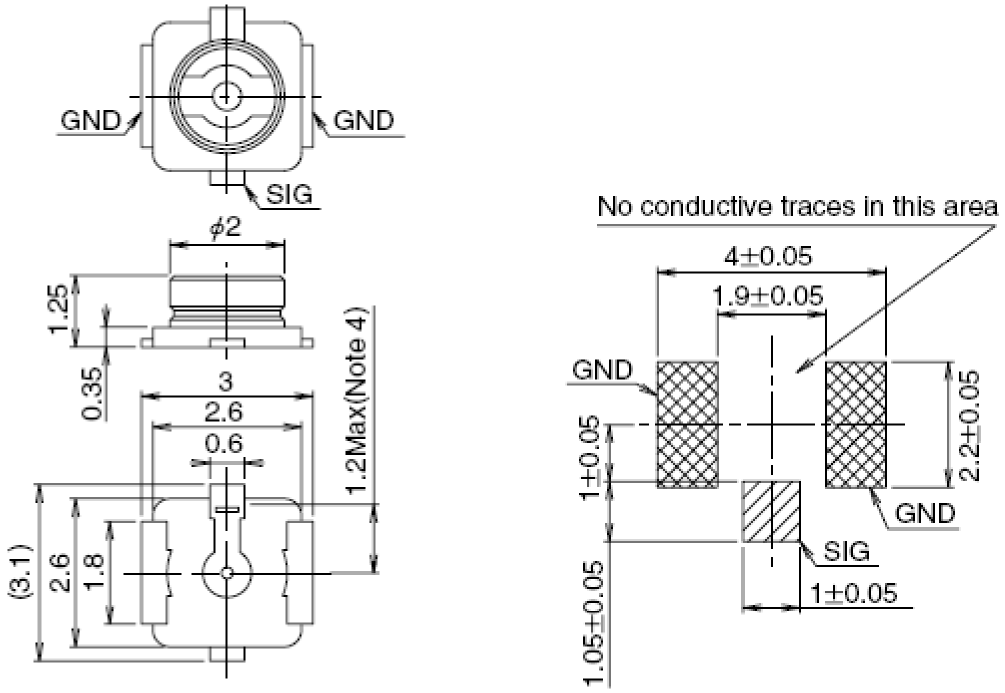
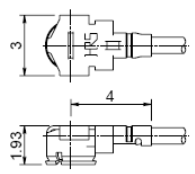
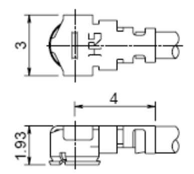
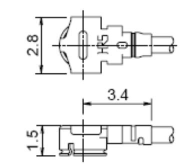
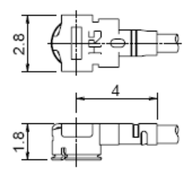
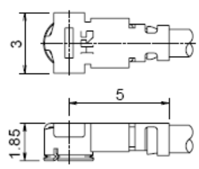


Figure 3: Antenna Connection Size (Unit: mm)

U.FL-LP series connecting wire can be used in conjunction with the antenna connector.

	U.FL-LP-040	U.FL-LP-066	U.FL-LP(V)-040	U.FL-LP-062	U.FL-LP-088
Part No.					
Mated Height	2.5mm Max. (2.4mm Nom.)	2.5mm Max. (2.4mm Nom.)	2.0mm Max. (1.9mm Nom.)	2.4mm Max. (2.3mm Nom.)	2.4mm Max. (2.3mm Nom.)
Applicable cable	Dia. 0.81mm Coaxial cable	Dia. 1.13mm and Dia. 1.32mm Coaxial cable	Dia. 0.81mm Coaxial cable	Dia. 1mm Coaxial cable	Dia. 1.37mm Coaxial cable
Weight (mg)	53.7	59.1	34.8	45.5	71.7
RoHS	YES				

Fiture 4: U.FL-LP Cable series

The following figure shows the installation dimensions of the connecting wires and connectors:

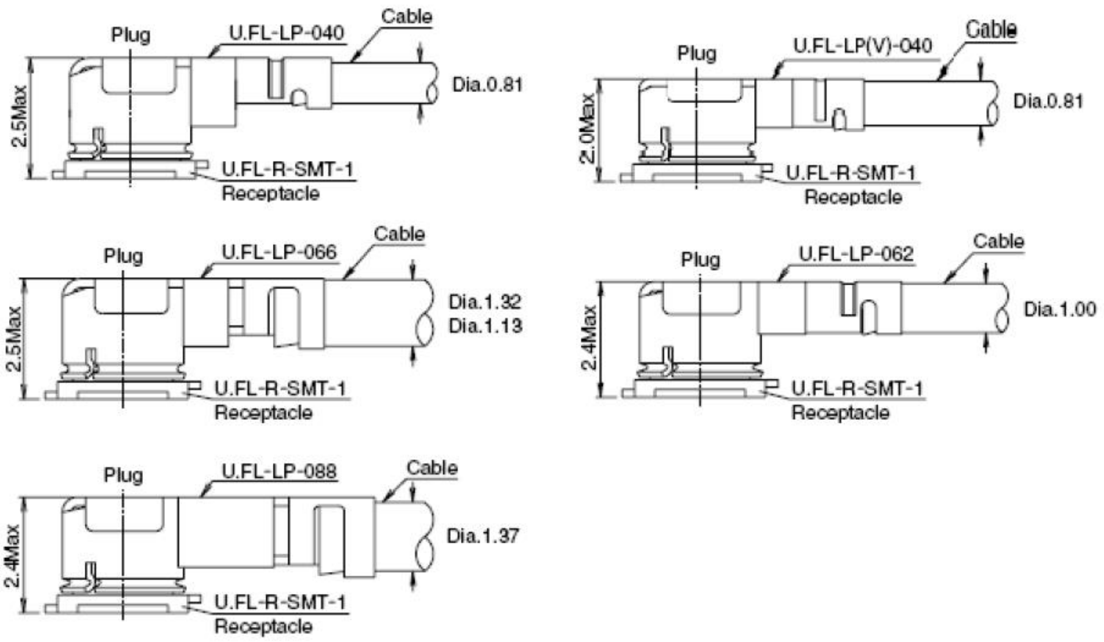


Figure 5: Installation size (Unit: mm)

6 Reliability and Electrical Characteristics

This chapter mainly introduces the electrical characteristics of the development board interface, including:

- Power supply characteristics
- Static electricity protection Power Supply Characteristics

The USB input voltage of the core board is 5.0~5.1 V, the pin input voltage is 5V-16V, and then converted from DCDC to 3.8V to supply the module. The power requirements are shown in the following table:

Table 6: Input Power Range

Param	Desc	Min	Typical	**Max**	**Unit**
VBUS	USB power supply	5	5.0	6	V
VIN	extern power	5	5.0	16	V

Table 7: I/O

Param	Desc	Min	Max	Unit
VIH	input high	0.7 × VCC	VCC + 0.3	V
VIL	input low	-0.3	0.3 × VCC	V
VOH	output high	VCC - 0.5	VCC	V
VOL	output low	0	0.4	V

*note: VCC typical value is 3.3V

6.1. Electrostatic Protection

Due to the static electricity generated by human body static electricity, charged friction between microelectronics, etc., static electricity can be discharged to the module through various pathways and may cause certain damage to the module. Therefore, static electricity protection should be taken seriously and reasonable static electricity protection measures should be taken. For example, wearing anti-static gloves during research and development, production, assembly, and testing processes; When designing products, add anti-static protection devices at circuit interfaces and other points that are susceptible to electrostatic discharge.
The following table shows the ESD withstand voltage of module pins.

Table 8: Electrostatic Protection

Test Interface	Contact discharge	air discharge	**Unit**
Power and ground interface	±4000	±8000	V
Antenna interface	±4000	±8000	V
other	±500	±1000	V

7 Notification

When using the core board, please pay attention to the following matters.

7.1. Spray

If spraying is required on the core board, please ensure that the spraying material used does not react chemically with the module shielding cover or PCB, and also ensure that the spraying material does not flow into the interior of the module.

7.2. Clean

Do not perform ultrasonic cleaning on the communication module mounted on the core board, as it may cause damage to the internal crystal of the module.

7.3. Turn-on

Short circuiting the BG or EG automatic power on pin can achieve automatic power on of the module when powered on. If power on/off control is required, the POWERKEY pin can be connected to an external control circuit.

Note

This manual does not represent any position or viewpoint of the company, and any losses resulting from misoperation guided by this manual are not related to the products of the mobile module.

8 Appendix Schematics

8.1. Reference Documents

Table 9: Reference Documents

Documents
[1] QuecPython-PH-7C4-P02_Core_Board_Specification_and_User_Guide_V1.0.0
[2] QuecPython-PH-7C4-P02_Schematic_Diagram
[3] QuecPython-PH-7C4-P02_Silkscreen